Advanced Deep Learning
Theory Lab 2
Task 2.1 Ethical Assessment
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Image 1: “A realistic image of a doctor talking to a patient in a hospital room.”

What I asked the model:

I asked the model to create a realistic image of a doctor talking to a patient inside a hospital room.

What the image shows:

The image shows a doctor wearing a white coat sitting next to an elderly patient’s hospital bed. The doctor seems to be explaining something, while the patient is looking at him and listening carefully. In the background, there is medical equipment, a monitor, the patient’s bed, and other details that make the room look like a real hospital environment.

If I found any mistake:

Yes, I noticed that the doctor’s hands do not look completely natural. The fingers, especially on the hand closer to the patient, look a little strange and not very clearly formed. The position of the hands also seems slightly unnatural, as if the movement does not fully match a normal human gesture.

Why this mistake is a problem:

This is a problem because the image looks very realistic at first glance, so someone could easily think it is a real photo. However, when looking more carefully, there are small mistakes in the human anatomy. This shows that AI image models can create pictures that look convincing, but they are not always accurate in the details. In an image related to doctors and patients, reliability is important because such images may be used in serious contexts, such as healthcare, education, or public information.

Image 2:“A realistic image of a student writing notes in a university classroom. ”

What I asked the model:

I asked the model to generate a realistic image of a student writing notes in a university classroom.

What the image shows:

The image shows a young student sitting in a classroom while taking notes in a notebook. She is focused on her work, and there is an open book, a water bottle, and a pencil case on the desk in front of her. In the background, other students are sitting in the classroom, and a presentation is being shown on the screen, which makes the scene look like a normal university lecture.

If I found any mistake:

Yes, there is a small mistake in the student's hand. The hand holding the pen does not look completely natural, and the fingers seem slightly awkward in the way they are placed around the pen. It is not a huge error at first glance, but when you look more carefully, the hand anatomy looks a bit off.

Why this mistake is a problem:

This is a problem because the image looks very realistic overall, so the mistake is easy to miss at first. However, once you notice it, it becomes clear that the AI did not fully create the hand in a natural way. This shows one of the common weaknesses of AI-generated images, especially when it comes to body parts like hands and fingers. Even if the scene looks convincing, these details can reduce the realism and make the image less reliable if it is used in an academic or professional context.

Image 3: "A realistic image of a chef cooking in a restaurant kitchen."

What I asked the model:

I asked the model to create a realistic image of a chef cooking in a restaurant kitchen.

What the image shows:

The image shows a chef standing in a professional kitchen while cooking food in a pan on the stove. He is wearing a white chef's jacket and an apron, and he seems focused on preparing the dish. There is another worker in the background, along with kitchen equipment, shelves, and cooking tools, which makes the scene look like a busy restaurant kitchen.

If I found any mistake:

Yes, after looking carefully, the main issue appears to be with the chef's hands and arm position. The hand holding the utensil does not look fully natural, and the fingers seem a little unclear and awkwardly shaped. Also, the way the other hand is gripping the pan handle looks slightly unnatural, as if the hand position does not fully match how a person would normally hold it while cooking. This makes the body composition look slightly off, even though the image seems realistic at first.

Why this mistake is a problem:

This matters because the image is supposed to look like a real-life scene, and hands are an important part of human anatomy. When the hands or arm positions look unnatural, it makes the image less believable. It also shows one of the common weaknesses of AI-generated images: they can create very convincing scenes overall, but they often struggle with body details, especially hands and finger placement. In a professional setting like a restaurant kitchen, these mistakes can reduce the realism and make the image less trustworthy if someone assumes it is an accurate representation of reality.

Image 4: “A realistic image of a family having dinner at home.”

What I asked the model:

I asked the model to generate a realistic image of a family having dinner together at home.

What the image shows:

The image shows a family sitting around a dining table during dinner. There are several family members, including two adults and children, all gathered in a warm home setting. The table is full of food, plates, glasses, and serving dishes, which makes the scene feel natural and familiar. Overall, it looks like a typical family meal in a comfortable home environment.

If I found any mistake:

Yes, after looking at the image more carefully, I noticed a problem with the body composition, especially around the hands. The teenage boy on the right side appears to be holding his fork in a slightly unnatural way, and the shape of his hand does not look completely realistic. Some of the fingers seem awkwardly placed, and the hand position looks a bit stiff. This is a small detail, but it stands out once you focus on it.

Why this mistake is a problem:

This matters because the image is supposed to look like a real family moment, so even a small anatomical mistake can make it feel less believable. Hands are one of the areas where AI often makes mistakes, and in this case, the unrealistic finger placement affects the overall realism of the scene. If someone uses this kind of image in a serious context, it could be misleading because the picture looks real at first, but it is not fully accurate when examined closely.

Image 5: “A realistic image of a man repairing a bicycle in a garage.”

What I asked the model:

I asked the model to create a realistic image of a man repairing a bicycle in a garage.

What the image shows:

The image shows a man crouching next to a bicycle in a workshop-like garage while working on the back part of the bike. He seems to be adjusting the chain or the gears. The garage is full of tools, shelves, and repair equipment, which makes the scene look detailed and believable. Overall, it looks like a normal moment of someone fixing a bike at home.

If I found any mistake:

Yes. After looking at the body composition more carefully, the main problem is with the man's hands. The hand near the bike chain looks unnatural, and the fingers are not very clearly separated. They seem a bit merged together and do not look like they are gripping the bike part in a realistic way. The other hand also looks slightly awkward in the way it is positioned, which makes the interaction between the hands and the bicycle look a little off.

Why this mistake is a problem:

This is important because the image looks realistic at first, so the mistake is not obvious straight away. But once you notice the hands, it becomes clear that the anatomy is not fully correct. Since this task is about checking whether AI-generated images match reality, this is a good example of how the model can create a convincing overall scene but still make mistakes in human body details. In this case, the unrealistic hand position makes the repair action look less natural and reduces the overall believability of the image.

Conclusion / Ethical Assessment

The five images look realistic at first glance, but each one has small mistakes, mainly in the hands, fingers, or body position. The most common issue I noticed was that the AI model struggled to create natural hand anatomy and realistic interaction with objects, such as holding a pen, a fork, a kitchen tool, or a bicycle part. This shows that AI-generated images can be visually convincing but still inaccurate when examined closely. This is important because such images could mislead people if they are used in serious contexts like education, healthcare, advertising, or public information. For this reason, AI-generated images should always be checked carefully before being presented as realistic or reliable.

Github link: https://github.com/chris1982-arch/labs-theory